

# Content-Preserving Motion Stylization using Variational Autoencoder

Chen-Chieh Liao<sup>†</sup>  
Tokyo Institute of  
Technology  
Tokyo, Japan  
liao.c.aa@m.titech.ac.jp

Jong-Hwan Kim  
NAVER Cloud  
Seongnam, South Korea  
jhwon.kim@navercorp.com

Hideki Koike  
Tokyo Institute of  
Technology  
Tokyo, Japan  
koike@c.titech.ac.jp

Dong-Hyun Hwang  
NAVER Cloud  
Seongnam, South Korea  
dh.hwang@navercorp.com

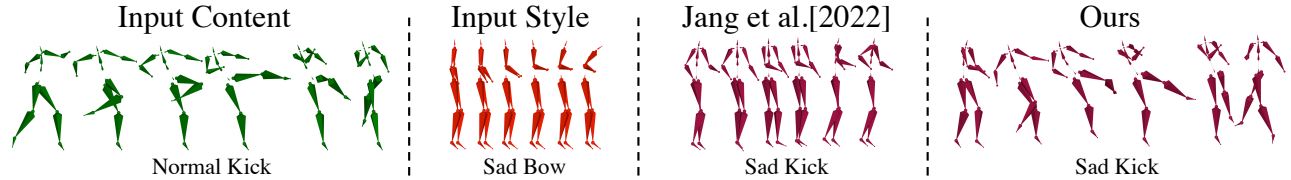


Figure 1: Our proposed style-transfer method can transfer the style (sad bowing) while well preserving the content of the input motion (neutral kick)

## ABSTRACT

This work proposes a motion style transfer network that transfers motion style between different motion categories using variational autoencoders. The proposed network effectively transfers style among various motion categories and can create stylized motion unseen in the dataset. The network contains a content-conditioned module to preserve the characteristic of the content motion, which is important for real applications. We implement the network with variational autoencoders, which enable us to control the intensity of the style and mix different styles to enrich the motion diversity.

## CCS CONCEPTS

• Computing methodologies → Machine learning.

## KEYWORDS

machine learning, motion style transfer, computer graphics

### ACM Reference Format:

Chen-Chieh Liao, Jong-Hwan Kim, Hideki Koike, and Dong-Hyun Hwang. 2023. Content-Preserving Motion Stylization using Variational Autoencoder. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603679>

## 1 INTRODUCTION

Metaverse has recently been highly noticed by the public, as many corporations are making virtual platforms, 3D characters, and animations. To control virtual characters and create various motions, characters' postures are often determined explicitly by animators

<sup>†</sup>This work was done when the author was on an internship at NAVER Cloud Corp.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603679>

and artists. On the other hand, to reduce the workload of creating precise motion poses frame by frame, motion capture systems have been developed to capture actors' motion, which can then be retargeted to virtual characters. However, the execution of motion capture and post-processing is still time-consuming and often requires the assistance of technical experts.

Recent works have focused on modifying one motion to another that performs the same motion but in different styles representing human characteristics such as emotions and states to increase the variety of motions. This so-called motion stylization is an active research field in the motion domain, and its objective is to reflect a style from one arbitrary motion to another. In previous studies [Aberman et al. 2020b; Jang et al. 2022], it is found that motion styles can be well transferred within the same motion category, such as walking and running. However, motion contents are hard to preserve when training with datasets that have a great variety of motions. These limitations make motion style transfer hard to be generalized to actual applications because one has to train networks for each motion category, which is inefficient. Furthermore, suppose the network can only work well in a particular motion category. In that case, it cannot benefit from transferring styles that do not exist in the training dataset from other motion categories.

To handle the above issue in motion stylization, in this work, we propose a motion style transfer method that works effectively on transferring motion styles among different motion contents. The network contains a content-conditioned module that takes account of the motion content while transferring the style to the target motion. In addition, the network is designed using variational autoencoders (VAEs), which enable the network to balance the constraints between contents and styles and to interpolate motion between styles in the latent space.

The contributions of this work can be summarized as follow:

- A generic approach for motion style transfer, suitable for modifying motion with styles from various motion contents
- The first motion style transfer network with VAE controlling latent content and style space
- A network can sample and control stylized motions that are unseen in the motion dataset

## 2 METHOD

In this work, we propose a motion style transfer network (Figure 2) using VAEs. This network takes two motion inputs: one is content motion and another is style motion, using the 6D rotation representation [Zhou et al. 2019]. The network has a two-layer structure using skeletal convolution and pooling [Aberman et al. 2020a] to encode the motions to the latent vectors.

As highlighted in Figure 2, the network comprises two latent spaces for content and style, respectively. Since the two inputs may come from totally different motions, transferring style between arbitrary motion pairs is challenging. Therefore, we use these two latent spaces to encourage style and content constraints during training that ensure the preservation of both the content and style. In addition, for real applications and industry-level usages, we want to have the ability to control the intensity of the style or combine different styles to form a unique motion that has been unseen before. For that, we use latent spaces to apply manipulations of motions for real applications and the expansion of existing databases.

Furthermore, inspired by a previous image-to-image work [Saito et al. 2020], we introduce a content-conditioned module when mixing the content motion with the style code. This module encourages the network to form a unique style being aware of the content feature, which helps preserve the motion’s content structure.

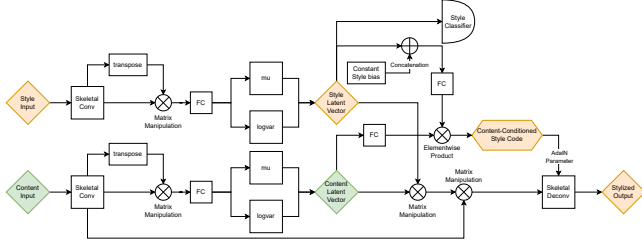


Figure 2: Overview of the proposed network.

## 3 EVALUATION

To evaluate the validity of the proposed method on motion stylization, we compare our method with the state-of-the-art method [Jang et al. 2022]. We train the networks with two motion datasets: the Xia dataset [Xia et al. 2015] cleaned by [Aberman et al. 2020b], and the Bandai-Namco-Research-Motiondataset [Kobayashi et al. 2023]. Both datasets are introduced for motion style transfer, while the Bandai-Namco dataset contains more challenging tasks, such as roundhouse kicking and long-term dancing.

As shown in Figure 1, we observe that in cases where the style motion has different content from the content motion, the previous work tends to modify the content motion to another motion that shares the same content with the style motion. On the other hand, the proposed method can address motion stylization while preventing changing the input content.

Furthermore, we have investigated quantitatively the capability of the network to stylize motion while preserving the content and style information. We evaluate the content consistency (CC) and style consistency (SC) by calculating the average Euclidean distance between the input and output motions. If the same style is shared

Table 1: Content consistency (CC) and style consistency (SC) comparison between the proposed method and the state-of-the-art using two different datasets.

| Dataset     | Xia              |                  | Bandai-Namco     |                  |
|-------------|------------------|------------------|------------------|------------------|
| Metric      | CC↓              | SC↓              | CC↓              | SC↓              |
| Jang et al. | .74 ± .37        | .31 ± .34        | .89 ± .62        | .46 ± .29        |
| <b>Ours</b> | <b>.20 ± .35</b> | <b>.29 ± .36</b> | <b>.36 ± .31</b> | <b>.53 ± .44</b> |

by both the input content and style motion, the output result is more reasonable to be close to the content motion. Therefore, when calculating the content consistency, we use the motion-pair with the same style label; likewise, when evaluating the style consistency, we use the motion-pair with the same motion content. Table 1 shows the results of the quantitative study. We observe that the proposed method outperforms the state-of-the-art method regarding content consistency. Along with the qualitative evaluation, the results show the effectiveness of the network in modifying input motions with styles while being aware of both content and style features.

In addition, during the experiment, we also find that the network is robust for small datasets where only one motion exists for each content. With such a dataset, the network can still learn style features among various motion categories, yielding the potential for few-shot or zero-shot learning. In the future, we plan to conduct further studies examining the robustness of the network for different motion datasets with different sizes.

## 4 CONCLUSION

In this work, we propose the first motion style transfer network using variational autoencoder to control both the content and style latent space. The proposed network is able to learn and transfer different style features from various motions, and it has the ability to sample unique motions from the latent space that are unseen in the dataset. A comprehensive study is conducted to compare the proposed method to the state-of-the-art method using the latest motion dataset specified for motion style transfer tasks. The results have shown that the motion content structure is well preserved during motion stylization, which is appalusive for real implementation and usage for production.

## REFERENCES

- Kfir Aberman, Peizhuo Li, Sorkine-Hornung Olga, Dani Lischinski, Daniel Cohen-Or, and Baoquan Chen. 2020a. Skeleton-Aware Networks for Deep Motion Retargeting. *ACM Transactions on Graphics (TOG)* 39, 4 (2020), 62.
- Kfir Aberman, Yijia Weng, Dani Lischinski, Daniel Cohen-Or, and Baoquan Chen. 2020b. Unpaired motion style transfer from video to animation. *ACM Transactions on Graphics (TOG)* 39, 4 (2020), 64–1.
- Deok-Kyeong Jang, Soomin Park, and Sung-Hee Lee. 2022. Motion Puzzle: Arbitrary Motion Style Transfer by Body Part. *ACM Transactions on Graphics (TOG)* (2022).
- Makito Kobayashi, Chen-Chieh Liao, Keito Inoue, Sentaro Yojima, and Masafumi Takahashi. 2023. Motion Capture Dataset for Practical Use of AI-based Motion Editing and Stylization. arXiv:2306.08861
- Kuniaki Saito, Kate Saenko, and Ming-Yu Liu. 2020. *COCO-FUNIT: Few-Shot Unsupervised Image Translation with a Content Conditioned Style Encoder*. 382–398. [https://doi.org/10.1007/978-3-030-58580-8\\_23](https://doi.org/10.1007/978-3-030-58580-8_23)
- Shihong Xia, Congyi Wang, Jinxiang Chai, and Jessica Hodgins. 2015. Realtime Style Transfer for Unlabeled Heterogeneous Human Motion. *ACM Trans. Graph.* 34, 4 (jul 2015). <https://doi.org/10.1145/2766999>
- Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. 2019. On the Continuity of Rotation Representations in Neural Networks. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5738–5746. <https://doi.org/10.1109/CVPR.2019.00589>